



Sophistry on steroids? The ethics, epistemology and politics of persuasive AI

Robin McKenna^{1,2}

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Abstract

This paper examines the ethical, epistemological, and political implications of persuasive AI technologies. Recent research suggests that AI is roughly as persuasive as humans in many contexts. Should this concern us? I argue that, while some worries about persuasive AI may be overblown, we should be worried for a mix of ethical, epistemological and political reasons. Most centrally, we should be worried because persuasive AI may lead to a small number of powerful actors dominating what I call the “marketplace of arguments”—the set of arguments that provide the materials we use to discuss important moral, political and societal issues.

Keywords persuasion · propaganda · AI ethics · marketplace of ideas · influence

1 Introduction

While many find an image of humans as masters of their own fate attractive, most of us recognise that it is unrealistic. Our lives are shaped by myriad factors outwith our control, to such an extent that we often don’t deserve much credit for our successes, or indeed much censure for our failures. The same goes for our intellectual lives—what we believe, what we think, where we gather our evidence. However attractive one finds an ideal of intellectual independence, of thinking for oneself, of making up one’s own mind without pressure from external influences, it is difficult to deny that there are myriad external influences on what we think and believe.

That we are so dependent on others in our intellectual lives has been emphasised by philosophers, in particular social epistemologists, for decades (Hardwig 1985; Kitcher 1990; for more recent discussion see Goldberg 2010; 2021). It is only because we are dependent on others that we have a large amount of the knowledge that we have. But that doesn’t mean there aren’t also problems that arise from our dependence on others. One set of worries concerns external

influences that are overtly coercive or manipulative. While you can’t literally force someone to believe something or think in a particular way, you can manufacture conditions in which they are more likely to believe what you want them to believe. This is the aim of indoctrination, and of techniques of mass communication such as propaganda (Bernays 1928 is admirably frank on this score). There is a lot to be said about indoctrination and propaganda (for recent discussions see Benkler et al. 2018; Hyska 2023; Ranalli 2024; Stanley 2015; Tuttle Ross 2002). But it is presumably not in question that these overtly coercive or manipulative forms of influence are *bad*, and they are bad even if they are less effective than many seem to think (Mercier 2020).

What about other external influences? There are external influences that are not coercive or manipulative at all: people sometimes provide information with the simple intent to inform. Then there are external influences that are not overtly coercive or manipulative, but which may be coercive or manipulative in a more subtle way. Persuasion is a mode of influence that is meant to differ from, say, indoctrination precisely in that it is not overtly coercive or manipulative. But suspicion of persuasion, especially emotive and heavily rhetorical forms of it, has a venerable history in philosophy, stretching back to Plato (Dow 2019; McCoy 2007).

Some have worried that even rational forms of persuasion—persuasion by means of argument and reasons—can be coercive. In interpersonal contexts, you might worry that you cannot simply separate the persuasive power of

✉ Robin McKenna
r.j.mckenna@liverpool.ac.uk

¹ Department of Philosophy, University of Liverpool,
Liverpool, United Kingdom

² African Centre for Epistemology and Philosophy of Science,
University of Johannesburg, Johannesburg, South Africa

an argument from the persuasive power of the person giving it, especially in contexts with power imbalances (Davis 2017; Tsai 2014). In mass communication contexts, the worry is that those designing and implementing mass communication strategies are more interested in attaining their goal—increased uptake for whatever ideas they are trying to “sell”—than in the means by which they attain that goal (Hausman and Welch 2010). This is clearly a worry in the context of marketing, whether commercial or political (Beckman 2018). But it is also a worry in the context of science communication (McKenna *forthcoming*). The worry here is that scientists and science communicators are often more interested in shaping public attitudes towards science than in the means by which they shape those attitudes; perhaps a lack of openness about certain complexities, transparency about underlying processes, perhaps even downright dishonesty is justified if it results in a public that is more “pro-science” (John 2018). Public health and public health communication is particularly important here, as this is an area where buy-in from the public is required for the success of vaccination programmes and the like. It is tempting to focus on achieving the buy-in, and not to worry too much if it is achieved via means that are subtly coercive or manipulative (Brown *forthcoming*; Oxman et al. 2022; Rossi and Yudell 2012; Susser 2020).

This paper is not directly about these well-established worries, though they provide the background context for much of the discussion. It is about a new set of worries, arising from recent developments in artificial intelligence (AI). There has recently been considerable speculation regarding the potential of AI technologies to influence attitudes and behaviours, especially about politics (for a small selection see: Hui et al. 2023; Burtell and Woodside 2023; Durmus et al. 2024; Floridi 2024; Salvi et al. 2024). I will go into more detail about what, exactly, this might involve in the next section. For now, let me just talk about “AI persuasion” as a catch-all term for the use of AI technologies to persuade people of things—to change their minds, to adopt new behaviours, and so on.

I aim to strike a middle ground between two positions. One position, recently defended by Luciano Floridi (Floridi 2024), has it that “AI persuaders” (AI chatbots who try to persuade people of things) or AI-generated “persuasive content” will be significantly more persuasive than humans, with serious political and social problems likely as a result. The other position has it that AI won’t add much to the persuasive power of existing technologies for “mass persuasion”, and so by itself will not lead to any serious social and political problems. (I say “by itself” because you might think it will lead to serious problems in part due to misapprehensions about its persuasive power). While this more sceptical position has not (to my knowledge) been defended in print, it fits with a more general scepticism about the

transformative potential (whether for good or ill) of AI that is relatively common at this point in the development of these technologies.

I think that Floridi’s position is unnecessarily alarmist and not supported by the existing literature, or reasonable extrapolations from it. But I also think that there are genuine concerns about persuasive AI. As I will argue, the fundamental reason why we should be worried about AI persuasion is *not* that AI has, or soon will have, significantly more persuasive power than humans. The reason we should be worried is that any AI persuader or AI-generated persuasive content is (a) relatively easy to roll out en masse and at scale when (b) you have the resources (financial as well as technological) to create it in the first place. That is, we should be worried about AI persuasion for roughly the same reasons that many are worried that a tiny number of tech companies control large amounts of the information we see online, as well as the infrastructure required for us to share and access that information. The worry is not so much about the *quality* of AI-produced persuasive content as the *quantity* of it, not to mention who has the power to produce and disseminate it.

Before getting to why I think we should be concerned, though, I want to start by doing two things. First, I will survey the existing literature on persuasive AI and what is possible with existing technologies (§2). This survey is important because the arguments I will go on to discuss are all grounded in these technologies as they currently exist, not as they might exist in some imagined future. Second, I will say a bit more about how I understand persuasive AI (§3). I will then run through three arguments, all of which conclude that we should be worried about persuasive AI (§4–6). While I think some of these arguments are more persuasive than others, given my overall aims I’m not too worried if the reader disagrees with my assessment of their relative persuasive power. I will be happy if, by the end of this paper, the reader is convinced that we should be worried about persuasive AI even if there is no reason to think that AI has significantly greater persuasive power than human beings, and even if the realities of what is possible with AI do not match the current levels of hype.

2 Persuasive AI: an overview

What, exactly, do I mean by persuasive AI? I’ll discuss how to distinguish persuasion from other modes of influence in the next section, but for now we can work with this rough definition: “persuasive AI” refers to the use of AI technologies to influence beliefs, behaviours, or attitudes. This might include using AI to create a persuasive message (a piece of text that is designed to persuade anyone who reads it of something), creating AI chatbots that engage with humans with the aim of persuading them of things, or any other use

of AI with the intent to influence beliefs, behaviours, or attitudes.

There is a rapidly growing literature documenting the extent to which AI technologies are effective at influencing human beliefs, attitudes, and behaviours. Starting with AI influence more generally, Burtell and Woodside (2023) discuss existing uses of AI (both generative and predictive) to influence a range of human behaviours, including via the design of recommendation algorithms, which heavily influence consumer behaviour and information consumption online. They also speculate about the impact of near-future developments in AI, in particular the development of AI chatbots that are functionally indistinguishable from humans, and the role that AI might play in spreading misinformation online. In a less speculative vein, Goldstein et al. (2024) find that propaganda generated by GPT-3 (a large language model or LLM) is nearly as persuasive as selected examples of “real world” propaganda. It isn’t much of a stretch to conclude that AI-generated propaganda will soon, if it isn’t already, be as persuasive as human-generated propaganda campaigns (GPT-3 has already been superseded by more advanced models). This is supported by research by the AI company Anthropic, which finds that the persuasiveness of LLMs increases with the scale of the model, with higher-scale models (e.g., Claude 3, a LLM made by Anthropic) proving to be as persuasive as humans in certain contexts (Durmus et al. 2024).

But what do we know about AI persuasion in particular, as opposed to influence in general? Let me highlight two recent papers, which I take to be representative of the developing body of literature. The first is a paper by Hui et al. (2023), which examines AI’s persuasive capabilities on political issues. They find that GPT-3 (the same LLM that Goldstein et al. looked at) can create persuasive messages that are as persuasive as messages created by humans across a range of political and policy issues, including bans on assault weapons, carbon taxes, and paid parental-leave programmes. Moreover, they asked participants in their study to rate the persuasive messages they were given (participants were not informed whether the message was created by a human or AI). They found that AI-generated messages were described as “more evidence-based and well-reasoned,” whereas human-generated messages focus more on “experiences, stories, and vivid imagery” (p. 4). This is perhaps not surprising, and it is debateable whether it tells us that much about the persuasive “tactics” of AIs. After all, you could presumably create an LLM that generated persuasive messages that focused on experiences, stories, and vivid imagery if you wanted to. The “preference” for evidence-based persuasive content may just reflect the training data (the data the LLM was trained on) or the training process more generally.

The second is a paper by Salvi et al. (2024), which also looked at AI and political persuasion. They conducted an elaborate series of experiments featuring structured debates between humans and AI chatbots (chatbots powered by the LLM GPT-4). These debates were either human-on-human or human-on-chatbot and were about a wide range of moral, political, and social issues. There were, roughly speaking, two different experimental setups. The first, which was the “basic” format, was a straightforward debate, with neither “side” given information about their “opponent”. In this format, the researchers found that the chatbots were as persuasive as humans. However, in the second setup, where the chatbots were given basic demographic information about their human opponent, such as gender, age, ethnicity, education level, employment status, and political affiliation, they found that the chatbots were significantly more persuasive. The researchers conclude that “not only are LLMs able to effectively exploit personal information to tailor their arguments, but they succeed in doing so far more effectively than humans” (p. 3).

Let me draw some general points from all this. First, there is good evidence that persuasive AI technologies (typically, LLMs) are at least as persuasive as humans, and there is some evidence that the most cutting-edge models are more persuasive than humans when they are given relevant information about the target audience. That providing relevant data about the target audience makes AI technologies more persuasive is not surprising, given that successful persuasive strategies and messages are typically tailored to their audience (Cialdini 2001; Maio et al. 2019; O’Keefe 2016). LLMs are likely very good at tailoring due to their extensive training on large corpora that include numerous persuasive arguments, and those who have speculated about the persuasive power of AI have linked this to their ability to tailor their message for a target audience (Burtell and Woodside 2023; Floridi 2024).

Second, to say that AI technologies are at least—or perhaps a bit more—persuasive as humans is *not* to say that they are “hyper-persuasive” in the sense that Floridi is concerned with. Bai et al. (2023) make this point explicitly:

Much speculation regarding the future application of AI to politics assumes these technologies could influence humans’ political attitudes and behaviors. However, experimental research on political persuasion generally finds small effect sizes, and the effects of persuasive efforts by political campaigns are typically small or null. Further, political persuasion is complex, potentially drawing upon a number of skills, including perspective-taking, knowledge of the topic, logical reasoning, clarity of expression, and knowledge of effective interpersonal influence techniques, with success ultimately in the hands of the person receiving the persuasive appeal. Persuasion is also uniquely challenging

in highly polarized settings, such as the contemporary US (p. 3).

Persuasion is *difficult*. It is particularly difficult within the arena of politics, and the contemporary US is hardly unique in being a highly polarized setting within which to conduct political debate. To be sure, this is not to say that persuasion is *impossible*. The best available evidence suggests that it is possible to change people's attitudes and behaviours, but effects sizes are typically small, albeit statistically significant (Coppock 2022). If we are to believe the studies I have cited in this section, then AI technologies are likely to have a small but significant impact on people's attitudes and behaviours.

You might ask: what about future generations of LLMs? What about other future developments in AI? Maybe Floridi is wrong to worry about “hyper-persuasion” now, but he may be right to worry about it in the future. While I can't argue for this here, I want simply to note that there are good reasons for thinking that the “challenge” of influencing attitudes and behaviours, whether through persuasion or more coercive and manipulative tactics, is inherently difficult, so difficult that future technological developments, even developments that are hard to foresee right now, are unlikely to alleviate it. Persuasion is inherently challenging because humans are, by and large, *epistemically vigilant* (Mercier 2020; Mercier and Sperber 2017; Sperber et al. 2010). One of the reasons why mass persuasion techniques such as marketing and propaganda typically have small (if any) impacts on people's attitudes is that, for basic evolutionary reasons, we are primed to look for signs that someone is not trustworthy or does not have our best interests at heart. To be sure, humans can be tricked; we are not infallible. But we are far from the gullible creatures that many who are worried about the persuasive power of new technologies depict us as being (Mercier 2017). It is therefore far from obvious that AI persuasion will ever be significantly more effective than human persuasion. There may simply be limits on how *persuadable* humans are.

3 Some conceptual clarifications

We know that AI technologies can persuade, and that they are at least as good at it as humans (which is not to say they are great at it). But is AI persuasion something we should be worried about? Before I can answer this question, I need to clarify a few more things.

Persuasion can be roughly defined as a mode of influence over another's attitudes and behaviour that does not involve coercion, force, or manipulation (O'Keefe 2016, 4). Working this up into a full definition would require *inter alia* saying something more precise about how exactly persuasion differs from more coercive or

manipulative modes of influence, which is tricky because at least some forms of persuasion are arguably manipulative (Brown forthcoming; Marshall forthcomingb; Marshall forthcominga; Nettel and Roque 2012; Tsai 2014). Because addressing these issues would take us too far afield, and they don't play a role in the argument that follows, I will set them aside here.

One thing that is important for my purposes is that persuasion is *intentional*: if I influence your attitudes or behaviour by accident, I haven't persuaded you. Imagine someone who is extremely deferential to a political commentator about political matters. Learning that the commentator thinks that raising taxes will harm the economy is enough for them to think that raising taxes will harm the economy. The commentator exerts a lot of influence over this person's political attitudes, but it would only seem right to say that the commentator has persuaded them of something when they form those attitudes after reading something the commentator has written. Merely learning that the commentator thinks something, and shifting attitudes as a result, is a form of influence, but it is not persuasion.

At this point, a critical question arises for the concept of “AI persuasion”. It is, many think, reasonable to hold the view that AIs lack intentions, and more generally that they are incapable of having any intention-like attitudes because they lack any mental states (Freiman 2024). But if AIs lack intentions, it seems like they cannot persuade because, by definition, persuasion is an intentional act. (You could argue for a similar point by denying that AI agents can act, though this may be a little trickier). At this point, a few options present themselves. First, we could loosen the standard definition of persuasion so that persuasion is not, by definition, an intentional act. Second, we could take on the—admittedly rather difficult—task of arguing that AIs can have mental states in general, and intentions in particular. Neither of these options strike me as particularly promising, and at any rate I won't pursue them here.

Luckily, there is a third option: we could talk about AI as a “persuasive technology” rather than talk of AIs as “persuaders” (Floridi 2024). The basic idea is that, following Floridi, certain technologies are technologies of persuasion: they provide ways of creating and transmitting persuasive messages. Viewed like this, the current discussion of the persuasive power of AI is in many respects similar to earlier discussions of the persuasive power of new media of mass communication, such as the radio, TV, and indeed the internet. Of course, all these technologies are used for many purposes besides persuasion. But the same is true of AI: it can be used for many things, and persuasion is but one of them. Crucially, while AIs cannot themselves have intentions, the purveyors of these persuasive AI technologies clearly can. Problem solved. (At least, solved in broad outline. Some fiddly details may remain, but the basic point should be clear).

Let's return to my question: should we be worried about AI as a persuasive technology? I want to set aside some worries you might have, not because they aren't important, but because they don't take much work to uncover. In particular, it would be relatively straightforward to argue that we should be worried about the potential for AI to be used to spread misinformation online. It is common to define misinformation in a fairly broad way, so that it includes content that is *misleading* as well as downright false or inaccurate (Roozenbeek and van der Linden 2024). This makes some sense: as philosophers have long recognised, deception is often more a matter of ensuring that someone (the target of the deception) forms false beliefs than saying things that are straightforwardly false (Carson 2009). Even if you think that worries about misinformation are often overstated (Winsberg forthcoming), it is plausible that recent developments in AI exacerbate them—or turn something that wasn't a problem into a potential problem.

I am interested instead in what you might call a “best case”—or at least a “better case”—scenario, in which the purveyors of persuasive AI technologies develop LLMs and chatbots that produce genuine arguments, evidence, and reasons for conclusions that the purveyors of these technologies themselves endorse, or at least would like their intended targets to endorse. That is, they produce genuine arguments with the attempt to *rationaly* persuade, not simply deceive or mislead.

In a recent paper, Thomas Mitchell and Thomas Douglas provide a useful definition of rational persuasion:

A rationally persuades B to adopt attitude α if: (i) A brings it about that B adopts α , (ii) A does so only by providing B with reasons for adopting α , (iii) B adopts α based on recognizing some of the reasons given by A, (iv) A intends each of (i)–(iii) (Mitchell, et al., 2024, 3).

This definition reflects a particular way of thinking about rationality and reasons. Because I will also assume this way of thinking, I need to say something about it. The basic idea is simple enough: what is distinctive about rational modes of persuasion is that they involve providing *reasons* for adopting a new attitude or changing one's existing attitudes. In their paper Mitchell and Douglas specify that by “reason” they mean a fact that counts in favour of (or against) the attitude in question ((Durmus 2024), 5). For example, the fact that it will reduce rates of transmission is a reason to support a vaccination programme, whereas the fact that it will have serious side-effects for many who take it is a reason not to support it, or at least to only support a limited rollout of it. While I will follow Mitchell and Douglas in assuming that reasons are facts, someone who endorses a different view, e.g. a view on which reasons are mental states as in Turri (2009), can simply replace my view with their favoured view where relevant.

Mitchell and Douglas's view of rational persuasion goes naturally with a way of thinking about rationality, on which having rational attitudes is a matter of responding appropriately to the reasons one has. That is, what makes rational persuasion *rational* is not just that it consists in providing reasons, but that someone who adopts or updates their attitudes because of those reasons thereby has a rational attitude. Note that by “rational” I mean *epistemic* rationality, as opposed to instrumental rationality, or practical rationality more generally. I am working with a very minimal conception of epistemic rationality here, on which it is a matter of having beliefs—more generally, attitudes—that are appropriately responsive to evidence, facts and reasons. This conception of (epistemic) rationality is minimal in the sense that it is consistent with various different substantive accounts, including the various forms of evidentialism (Conee and Feldman 2004) and reliabilism (Goldman 1979). (It should be obvious that it is consistent with evidentialism. It is consistent with reliabilism if we avoid the assumption that appropriately responding to evidence or facts requires the conscious awareness that one is appropriately responding).

To use this definition of rational persuasion to think about AI persuasion, we need to modify it a bit, in line with our earlier discussion of the sense in which AI technologies can be said to persuade. Properly speaking, we shouldn't say that an AI (rationally) persuades someone to adopt an attitude. We should say that the purveyor of the AI technology does this. So perhaps we can say this: the purveyor of a persuasive AI technology rationally persuades someone to adopt an attitude when the purveyor uses the technology to bring it about that the intended target(s) adopt the attitude, the purveyor does this by using the technology to provide the target with reasons for adopting the attitude, the target adopts the attitude on the basis of these reasons, and the purveyor of the technology intends that all of this be the case. There may be some issues in cases where the purveyor of the persuasive AI technology does not know what the attitude in question is (or what the reasons the technology provides are), but we could say that, in these cases, it is still the case that the purveyor of the technology intends that the intended target adopt *an* attitude on the basis of *some* reasons that the technology provides.

My questions then are the following: what, if anything, is wrong with using persuasive AI technologies to rationally persuade people to change their attitudes and behaviours? For example, what would be wrong with using these technologies to rationally persuade people to be more worried about climate change, more supportive of mass vaccination programmes, or providing foreign aid? (I have chosen examples where my reader may well think that it is good for people to believe these things; I don't want to make things too easy for myself). On the other side of the equation, what, if anything, is wrong with being persuaded by the outputs of

persuasive AI technologies, especially when those outputs take the form of reasons and arguments? For example, would it somehow be irrational to be persuaded by the outputs of these technologies?

In the next few sections, I consider three arguments that we should be worried about persuasive AI. Two of them are intended to show that we should be worried about the use of persuasive AI technologies on ethical and political grounds (54–6). The other argument is intended to show that we should view the attitudes formed in response to the content produced by persuasive AI technologies as irrational, or at least as less rational than they would be if they had been formed via some other means not involving AI (§4). I am more impressed with the ethical/political arguments than the argument concerning rationality. But I don't mind too much if the reader disagrees, given that my overall aim is to argue that we should be worried about AI persuasion.

4 The rationality argument

The Rationality Argument says that we should be worried about persuasive AI technologies because they would undermine the rationality of our attitudes, beliefs or credences. That is, if we are persuaded to change our attitudes, adopt a new belief, abandon an old one, or update our credences by persuasive AI technology, then that change of attitude/belief/credence is irrational, or at least less rational than if it had been brought about by other means, for example, via persuasive messages and strategies devised by humans, or via thinking about the issue(s) for oneself. Bear in mind that by “rational” I mean the minimal sense of epistemic rationality explained in the previous section: an attitude is epistemically rational just in case it is an appropriate response to the relevant evidence, facts, and reasons. The worry then is that, when we form or modify our attitudes via interactions with persuasive AI technologies, we end up with attitudes that are not an appropriate response to the relevant evidence, facts, and reasons.

There is a simple version of the Rationality Argument that can be dismissed quickly. The simple version says that it is simply irrational to be persuaded to change your attitudes/beliefs/credences by persuasive AI technologies. The problem with this is that, once we specify that we are focusing on outputs of persuasive AI technologies that take the form of arguments and reasons, it is hard to see why this would be so. Imagine someone creates an AI chatbot that provides lots of good arguments for reducing the burden of taxation in general, and specifically on the very rich. Someone interacts with this chatbot, thinks about the arguments, and is persuaded to become substantially less confident that taxes should be increased for the rich than they were before. What-ever you think about the *truth* of these claims (as indexed to

the tax system of a particular country), it is hard to see what would be *irrational* about responding in this way. This is simply a case of updating attitudes based on new evidence (the arguments that were given).

You might try to buttress the simple version of the Rationality Argument by highlighting the role that perceived source credibility plays in rational assessment of arguments. There are some situations where you can assess an argument independently of knowing anything about the source. Some arguments don't appeal to any empirical claims but rest on logical or conceptual points that can, at least with requisite training, be grasped merely on the basis of understanding their contents. Some arguments appeal to empirical claims, but the person assessing them has the relevant empirical knowledge to assess those claims. But, a lot of the time, the best way to evaluate an argument, specifically the empirical claims embedded within it, or on which it relies, is by assessing the credibility of the person giving the argument. Someone gives you an argument for reducing current rates of taxation, citing data about the likely impact on growth, GDP, etc. Unless you have the facility to evaluate the data for yourself, your assessment of this argument will largely be based on your assessment of the person giving it—do you trust them to accurately cite data, to use data in ways that don't prejudge the questions at issue, and so on?

This raises some problems when applied to persuasive AI technologies, especially if you favour frameworks for thinking about source credibility that foreground interpersonal relationships and other distinctively human (or at least agentic) features of the source in ways that don't obviously work when applied to AI, even “agentic AI” (Hinchman 2005; Moran 2005; Ross 1986). There are some frameworks for thinking about trust in AI that promise to resolve these problems (Carter 2023; Simion and Kelp 2020; Song 2023). At any rate, the basic problem with this buttressed version of the Rationality Argument is that there is no reason to think that we should trust AI-generated content or AI chatbots less than human-generated content or humans more generally. This is especially so if, as before, we bracket concerns about the use of persuasive AI technologies by bad actors. To be sure, there are reasons to worry about the reliability of AIs—their tendency to hallucinate, errors in the training data, biases in the training process, and so on. But, clearly, there are also reasons to worry about the reliability of humans and human-created persuasive content—humans lie, humans get things wrong, humans obfuscate, and so on. Absent reason to think that AIs are typically (far) less trustworthy and reliable than humans, the argument doesn't get off the ground.

A more nuanced version of the Rationality Argument would say that, while it may be rational to be persuaded to change your attitudes/beliefs/credences by persuasive AI technologies, this change in attitude/belief/credence is in some way *less* rational than if it had been brought about

by other means. More generally, the problem with persuasive AI technologies would then be not that interacting with them renders our attitudes irrational, but that interacting with them simply renders our attitudes less rational than they otherwise would be.

The challenge though is to give some reasons for thinking that forming or changing our attitudes by interacting with persuasive AI technologies will lead us to have attitudes that are less rational than they otherwise would be. One alternative to forming attitudes via interacting with persuasive AI technologies would be by forming attitudes without any outside influences: simply think about the issues for yourself, with no or minimal assistance from others. The problem with this, though, is that not only is it unrealistic to expect people to think about most issues without much assistance from others, it is also unclear why it would be a good thing to eschew any form of intellectual dependence on others (Levy 2024). Indeed, those who have defended intellectual autonomy (thinking for oneself) in the recent literature tend to argue that genuine autonomy involves “wise” or “discerning” forms of deference, rather than intellectual independence (Carter 2020; Roberts and Wood 2007; Zagzebski 1996).

Moreover, even if it is sometimes possible and desirable to form attitudes by yourself, without any external influences, it isn’t clear that the language of rationality captures what is desirable about it. At least as I am understanding it, rationality is a function of how well one responds to evidence and reasons, not of where one obtained one’s evidence or reasons. If you go out and obtain your own evidence, or do the work of sifting through the various pieces of evidence yourself, you may end up with a deeper level of understanding of the relevant issues (Kvanvig 2003; Matheson 2022; Pritchard 2016). But it is unclear why your attitudes will necessarily be any more rational than if you had simply updated them based on the evidence at your disposal, no matter where that evidence came from.

The alternative to forming attitudes via interacting with persuasive AI would be forming attitudes by interacting with other people. (Anyone who interacts with the various information sources and networks we have constructed is likely interacting both with people and with AI, so this isn’t a further, separate possibility). While it is plausible that some interpersonal interactions are better—from the standpoint of forming rational attitudes—than any interaction with AI technologies could be, it is equally plausible that other interpersonal interactions are worse from this standpoint. Having a discussion with your incredibly well-informed and trustworthy friend about the fall of the Gaddafi regime in Libya is a great way of forming rational beliefs about the causes, specifics and consequences of the fall of the Gaddafi regime. Because LLMs (put roughly) provide summaries of known information about a huge range of topics (or at least

appear to—they can be inaccurate), they lack the specificity and unusually high quality of the information you might get from your well-informed and trustworthy friend. On the other hand, having a discussion with your incredibly ill-informed and untrustworthy friend about the same topic is not a good way of forming rational beliefs about the topic. You would do better to rely on the summary provided by the LLM. So the best we can say is that interacting with persuasive AI technologies would not be an adequate substitute for interacting with someone who is incredibly well-informed and trustworthy about whatever topic you want to discuss. But this is hardly a major shortcoming of persuasive AI technologies, given that people who are incredibly well-informed and trustworthy about a huge range of topics are in quite short supply.

5 The sophistry argument

The Sophistry Argument starts from the thought that there is a crucial difference between human and AI persuaders. When a human tries to persuade another human of something, they have some “skin in the game”, so to speak. Either they are trying to persuade someone else of something they believe, in which case they have skin in the game in the sense that they are committed to the truth of the relevant claims and ideas, or they are trying to persuade someone else of something they are—for whatever reason—pretending to believe, in which case they have skin in the game in the sense that they are committed to it looking like they are committed to the truth of the relevant claims. (It is perhaps technically possible to try and persuade someone of something while openly acknowledging you don’t believe it, but this is unlikely to be an effective tactic). In contrast, “AI persuaders” lack any skin in the game for the simple reason that AIs—let’s assume again—lack beliefs (and skin), or indeed any other mental states. As a consequence, they can’t be committed to the truth of the claims and ideas for which they might present arguments, or engage in any sort of pretence that they are committed to them.

To make this argument work, though, some reasons need to be given for thinking that lacking any skin in the game is a bad thing. Indeed, you might think that lacking any skin in the game is sometimes a *good* thing. It is precisely because human persuaders have skin in the game that they sometimes engage in lies, deceit and other forms of subterfuge to bring it about that other people believe things that will serve their (the persuaders’) interests. Of course, the purveyors of persuasive AI technologies can also engage in lies, deceit and subterfuge, so this might not be a point in favour of persuasive AI. But, equally, it doesn’t seem to be a point against it.

One initially promising idea is to view AI persuaders as engaging in something akin to Frankfurtian “bullshit”

(Fisher 2024; Hicks et al. 2024). For Frankfurt, the bullshitter differs from the liar because, where the liar is invested in the truth because they want to conceal it, the bullshitter is indifferent to what is true or false (Frankfurt 1986). The reasons behind this indifference may vary. The bullshitter may simply want to distract their audience, perhaps by diverting their attention elsewhere. The purpose of their bullshit is therefore distractive rather than persuasive. But bullshit can also serve more persuasive purposes. The politician who makes claims without regard to their truth may be doing so to signal their commitment to a set of values communicated by the claims. They don't necessarily want to persuade their audience of the truth of the claims, but they very much want to persuade their audience that they are committed to the values that the claims convey.

While it may be tempting to view AI persuasion as akin to bullshitting, the problem with doing so is that bullshitting requires communicative intentions, whether to distract or persuade. If we assume—again—that AIs lack mental states, an AI persuader cannot intend to distract or persuade their audience of anything. Moreover, an AI persuader cannot be indifferent towards the truth in anything other than the literal sense that they have no attitude towards it. We could try to make the move we made earlier and say that, while AIs lack intentions, the purveyors of AI technologies don't. But it is unclear whether the purveyor of AI technologies can use them with the intent to distract or persuade their users in the way that the bullshitter does.

A better idea—I will argue—is to view AI persuaders as akin to sophists. The Sophists in ancient Greece were teachers of argumentation, philosophy, and rhetoric who made a living selling certain skills. While it may be a bit unfair to the ancient Sophists (Barney 2006), in ordinary language, the words “sophist” and “sophistry” carry pejorative connotations: a sophist uses clever, but perhaps misleading, arguments to win a debate, and sophistry is the art of using such arguments to win debates. A sophist is, if you like, an “arguer for hire”—they will tell you how to put together a cogent argument for a conclusion, irrespective of whether they endorse the conclusion, or the premises they suggest you use to argue for it. Indeed, if you want, they can give you another argument for a completely different conclusion, one diametrically opposed to the first one. The suggestion then is that we view AI persuaders as roughly analogous to sophists. Just like the sophist, the AI persuader will give you an argument for a conclusion, or tell you how to construct one, without endorsing the conclusion. We can also view the persuasive messages you might produce using AI as analogous to what you would get if you paid someone to create some “persuasive content” for you. (You might not be paying the AI for their work, but you are paying the company that created it, or contributing in some less direct way to their financial success).

But what, exactly, is wrong with sophistry? One option would be to argue that sophistry is, in general, wrong, and so we should be worried about AI persuasion simply because it is a form of sophistry. Another option—the one I will pursue—is to argue that, even if there is nothing necessarily wrong with sophistry, there are reasons to be worried about persuasive AI technologies because of the *kind* of sophistry they might enable. They might enable what I call “sophistry on steroids”. Let me explain.

Here is a truism: in a huge number of areas of possible dispute, there are numerous good arguments on multiple sides of the issue. Perhaps there are some views, on some issues, for which not much can be said. Perhaps there are some issues where the correct view of the matter is obvious to anyone with even a minimal degree of open-mindedness. But either of these situations is rare: usually, there are good arguments for different, logically incompatible conclusions. Someone who wants to think about these issues, or at least think about them responsibly, is going to want a way of getting access to all these arguments. Because any one person is going to struggle to come up with all the arguments for themselves, we are inevitably going to rely on each other for the production of arguments. The “quality” of our thinking about any given issue is going to, in large part, depend on whether there are mechanisms for producing a good number of arguments on many sides of the issue, and for ensuring these arguments are disseminated widely enough that they inform our thinking (Mercier and Sperber 2017).

Following the common metaphor of the “marketplace of ideas”, it is helpful to think about this in terms of a “marketplace of arguments”: there is a need (for arguments) and so there will be suppliers of arguments who step up to meet this need (cf. Williams 2023 on the “marketplace of rationalisations”). In interpersonal interactions, these markets will likely be very informal: we're discussing some issue, and the discussion will go better if the discussants put forward a variety of arguments on different sides of the issue. To the extent the discussants are invested in the discussion, they will look to provide these arguments. In public discourse, things are a bit more formalised: there are institutions that, at least nominally, play the role of providing and disseminating arguments (media, think tanks, universities), and individuals, often members of these institutions, who construct and disseminate these arguments to the public.

Whether we're talking about interpersonal discussions or public discourse, the hope is that these marketplaces will supply a representative sample of arguments on different sides of the various issues. That is, whatever the “balance of arguments” would look like in logical space, the market provides a representative, albeit inevitably imperfect, set of these arguments: all the central arguments on all sides are included, along with proportionate numbers of less central or weaker arguments. To take a silly philosophical example:

if we're talking about the nature of morality, we would want the central arguments for the various positions (moral realism, moral anti-realism, moral scepticism, etc.) to be represented, and it would be a problem if only the arguments for (or against) one of these positions were available. Of course, actual discourses will often *not* meet these criteria for various reasons, and there is the deeper issue that opinions on whether any given discourse meets these criteria will inevitably vary (a moral realist will likely think the balance of arguments supports their position, while a moral anti-realist will likely take the opposite position).

Setting these complications aside, the point is this: Whatever issues exist with existing marketplaces for arguments—and there may be many—persuasive AI technologies have the potential to greatly exacerbate them. This is for two different but related reasons. The first is that persuasive AI technologies have the potential to dominate the marketplace of arguments. For humans, contributing to the marketplace of arguments is time-consuming and labour-intensive—it involves coming up with new arguments, or finding ways of recycling old ones, and then looking for a way of getting those arguments disseminated more widely (e.g. publishing them somewhere where people will see them). Persuasive AI technologies provide a way of avoiding these time and labour constraints. The purveyors of persuasive AI technologies can generate huge numbers of “arguers for hire” (AI agents) or huge amounts of “persuasive content” within minutes, with any dilution in the quality of arguments often more than made up for by the massive increase in quantity. The second reason is that not just anyone can create and run persuasive AI technologies. The resources—not just financial, but also in terms of infrastructure, skills, and knowledge—required to design and operate a LLM are massive, so much so that only a very small number of companies operate in this field, at least at the level relevant to the argument of this paper (e.g. Anthropic, OpenAI, DeepMind, Microsoft, Meta). This means that the barriers to entering the marketplace in this way are very high, so high that ordinary people cannot hope to meet them.

Putting these two reasons together, the problem with persuasive AI is that a very small number of actors have the power to dominate the marketplace of arguments. Even if these actors have “good intentions”, and design persuasive AI technologies in ways that they think will promote what they view as true conclusions, this creates the potential for significant imbalances in the marketplace, with this small number of actors having the power to determine the balance of arguments that gets represented on any given issue. If they don't always have good intentions, or are simply wrong about which arguments are cogent or which conclusions true, the problems get worse. Anyone who is worried about situations where marketplaces are dominated by a small number of actors with the power to crowd out alternatives

will see the problem here. (Anyone who is worried about marketplaces in the first place should view it as a problem that this is a helpful metaphor for viewing the situation). In short: persuasive AI may, as I argued earlier, be a form of sophistry—AI persuaders are, like the ancient Sophists, “arguers for hire”. But, if it is a form of sophistry, it's *sophistry on steroids*. The very small number of companies that operate these persuasive AI technologies have the power to create armies of “arguers for hire” at will, and so to distort the marketplace of arguments, shaping it towards their ends.

My main point in this section has been that the problem with persuasive AI is that it gives a small number of actors disproportionate power and influence over the pool of arguments available for public consumption and consideration. Let me finish by highlighting some further problems with persuasive AI that follow from this. It is a familiar point that, for all their transformative potential, markets often serve the material interests of some people better than others. Benefitting from—or even participating in—a market requires resources like time, information and skills. Some people have more of these resources than others, with the result that they gain far more from the market. Moreover, these differences in resources are not simply the result of random chance; they typically reflect existing inequalities within society (Herzog 2024). The concern, therefore, is that not only will persuasive AI distort the marketplace of arguments, but that it will do so in ways that reflect and reinforce existing inequalities and imbalances in power.

This concern goes beyond worries about AI and algorithmic bias (Fazelpour and Danks 2021). Persuasive AI technologies have the potential to influence and shape attitudes about a wide range of contentious political issues. This only exacerbates existing concerns about the outsized influence that big tech companies exert on political debate (Aytac 2025). It is hard to view today's social media sites (X/Twitter, Facebook, etc.) as genuinely democratic public spheres. It is even harder to see how we could create democratic public spaces online when you add persuasive AI technologies into the mix. The companies that create these technologies have their own incentives. In some cases, the incentives seem largely financial: they exist to make money. In other cases, it is a bit murkier. Whatever these incentives are, though, it is hard to credibly argue that they are aligned with the public good.

One place where this may be particularly troubling is when it comes to control over information. Even if you are sceptical of the idea that there is a “right to be informed” (Marciel 2023) or a “right to know” (Watson 2021), it is clearly true that, in a democratic society, information matters. Having the power to decide which information people have access to (or lack access to) is not quite the same as having the power to decide what people think; you can dictate what people see and hear, but not what they think.

Persuasive AI technologies threaten to close this gap. While I am sceptical that they will close it entirely (persuasion is difficult, even when powered by AI) I am very much concerned about the idea that big tech companies will increasingly have the means not just to control what information we have access to, but to control the marketplace of arguments.

6 The political argument

Finally, the Political Argument focuses on a problem that might arise with the implementation of any set of guidelines or regulations for the ethical use of persuasive AI technologies. My naming this argument “political” should not be taken to imply a distinction from the previous argument, which you might also view as political. I call it this simply because the argument is that any set of guidelines for the ethical use of persuasive AI technologies will almost inevitably become the target of political battles and disagreement.

Let me state the argument in very abstract terms. Imagine that, the other arguments in this paper notwithstanding, we reached an agreement on a set of conditions X, Y, Z for the permissible use of persuasive AI. The set of conditions will presumably reference various factors, including whether the technology is being used to persuade people of things that are true (or at least believed to be true), whether it is trying to persuade via valid arguments that use premises that are (believed to be) true, whether people are likely to be exposed to a balance of arguments on different sides, and whether the intentions of the purveyors of the technology are good, or at least not malicious.

The problem though is that, on a very wide range of issues—including many where persuasive AI technologies are likely to be deployed—there will be significant disagreements about whether these conditions are met, whether by particular persuasive AI technologies, or by particular uses of those technologies. For example, there is likely to be significant disagreement about whether persuasive AI technologies are being used to persuade people of things that are true because there is likely to be disagreement about what things are true. To be sure, there won't *always* be disagreement about what things are true; consensus is sometimes arrived at in even the most heated of disputes. It may also be that, sometimes, disagreement is intractable because one side is simply being unreasonable. But the point is just that there will *often* be disagreement about what things are true, not that there will always be disagreement, or that any disagreement there might be is reasonable. As a result, the conditions on the permissible use of persuasive AI technologies are unlikely to resolve *political* problems that might result from the widespread

deployment of persuasive AI technologies. Indeed, they are likely to create new problems, or at least new arenas within which existing political problems are debated and existing political battles fought.

Let me give an example to illustrate the sort of thing I have in mind (to avoid the example becoming dated, I'll not make it too specific). Imagine a new vaccine has been developed for a potentially serious illness and it has passed through the required regulatory procedures. Public health communicators are then tasked with getting buy-in from the public for the vaccine: for herd immunity, vaccination rates need to be very high, ideally above 95%. To this end, they utilise persuasive AI technologies, designing communications that highlight the benefits of getting vaccinated, and perhaps even designing AI chatbots, whether for the purpose of persuading the public at large or for persuading those who are hesitant to get the vaccination themselves or get their children vaccinated. The view you take of this particular use of persuasive AI will, of course, depend on the view you take about the vaccine: do the costs, such as they are (side effects etc.), outweigh the benefits, such as they are? Someone who is sceptical that the costs outweigh the benefit, whether for this vaccine or vaccines in general, is going to be very suspicious about this use of persuasive AI, and may end up campaigning vociferously against it. They are certainly not going to agree that this particular use of persuasive AI is legitimate. This is not necessarily because they will view any use of persuasive AI as legitimate; it is simply that, in their view, the conditions for legitimate use have not been met.

The more these situations arise—situations where the use of persuasive AI is seen to serve political ends that are viewed by some as illegitimate—the more suspicion of the underlying technology there will be. Ultimately, it is hard to see how persuasive AI technologies can avoid getting caught up in the political and cultural disagreements into which they might be brought as a means of adjudication. The result will be that they are not just useless attempts to arbitrate these disagreements but potential sites of new cultural and political battles.

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